Theorem. *A connected quiver Q is of finite type if and only if the corresponding unoriented graph (i.e., with directions of arrows forgotten) is a Dynkin diagram (see Theorem 6.5.2 below).*

In this problem you will prove the "only if" direction of this theorem (i.e., why other quivers are NOT of finite type).

(a) Suppose Q has n vertices and no self-loops. Show that for any rational numbers x_i, $1 \le i \le n$, which are not simultaneously zero, one has $q(x_1, \dots, x_n) > 0$, where

$$q(x_1, \dots, x_n) := \sum_{i \in D} x_i^2 - \sum_{i,j \in D} b_{ij} x_i x_j.$$

Hint: It suffices to check the result for integers: $x_i = m_i$. First assume that $m_i \ge 0$, and consider the space W of representations V of Q such that $\dim V_i = m_i$. Show that the group $\prod_i GL_{m_i}(k)$ acts with finitely many orbits on $W \oplus k$, and use Problem 6.1.2 to derive the inequality. Then deduce the result in the case when m_i are arbitrary integers.

(b) Deduce that q is a positive definite quadratic form.

Hint: Use the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$.

(c) Show that a quiver of finite type can have no self-loops. Then, using Problem 6.1.3, deduce the theorem.

Problem 6.1.6. Let $G \neq \{1\}$ be a finite subgroup of $SU(2)$ and let V be the 2-dimensional representation of G coming from its embedding into $SU(2)$. Let V_i, $i \in I$, be all the irreducible representations of G. Let r_{ij} be the multiplicity of V_i in $V \otimes V_j$.

(a) Show that $r_{ij} = r_{ji}$.

(b) The **McKay graph** of G, $M(G)$, is the graph whose vertices are labeled by $i \in I$, and i is connected to j by r_{ij} edges. Show that $M(G)$ is connected. (Use Problem 4.12.10.)

(c) Show that $M(G)$ is an affine Dynkin diagram (one of the "forbidden" graphs in Problem 6.1.3). For this, show that the matrix $a_{ij} = 2\delta_{ij} - r_{ij}$ is positive semidefinite but not definite, and use Problem 6.1.3.

Hint: Let $f = \sum x_i \chi_{V_i}$, where χ_{V_i} are the characters of V_i. Show directly that $((2 - \chi_V)f, f) \ge 0$. When is it equal to 0? Next, show